

Service now - ITOM

Configure an IP address range for the MID Server

Configure an IP address range for the MID Server

Before you begin

Role required: agent_admin or admin

About this task

Applications, such as Discovery and Orchestration, can specify an IP range or the specific IP address of a target. When the application looks for a MID Server to use during auto-selection, it chooses a MID Server whose IP range includes the application's range or specific IP address. Applications also use other criteria, such as the MID Server's supported application or capability.

Note: This is a manual configuration procedure. To learn how the system can assign IP address ranges to selected MID Servers automatically,

These applications use IP ranges as follows:

- **Orchestration:** Use the IP address of the target machine (together with the capability) to select the correct MID Server for Orchestration activities.
- **Service Mapping:** Select the MID Server whose IP address range matches the IP in the discovery request.
Discovery: Can select an IP address range using the same criteria as Service Mapping. Discovery can also use a [quick IP range](#) that you specify in the Discovery schedule.

Tip: You can also use the **ALL** IP range, which allows the MID Server to be used with any range of IP addresses that an application specifies. This only allows the MID Server to access IP ranges, it does not mean that the MID Server can actually reach all of the IP ranges. To function with an application, the MID server must have access to the IP ranges that the application needs.

An IP address in dotted decimal or hexadecimal format. The hexadecimal format can be explicitly prefixed with 0x, this is not mandatory. Here are examples:

- 10.11.144.155
- 0x0A0B909B
- 0A0B909B

An IP address range in dotted decimal or hexadecimal format.

Here are examples:

- 10.11.144.150-10.11.144.160
- 0x0A0B9096-0x0A0B90A0
- 10.11.144.150-0x0A0B90A0

An IP network address with the net mask specified after a slash (/) in regular notation (0-32 inclusive) or in IP address notation. Here are examples:

- 10.11.144.0/24 10.11.144.0/255.255.255.0
- 10.11.144.0/0xFFFFFFFF00
- 0x0A0B9000/24
- 0x0A0B9000/0xFFFFFFFF00

Procedure

1. Navigate to **All > MID Server > IP Ranges**.
2. Click **New**.
3. Complete the form, using the fields in the table.

IP address ranges	
Field	Description
Name	Enter a descriptive name.
Application	Application scope for this IP range. The default is Global.
Type	Select the type of range: • Include: Include the specified IP range. • Exclude: Exclude the IP address range. Note: If a MID Server is configured for ALL applications, any IP addresses configured for exclusion are ignored by the system. If your goal is to configure default MID Servers for separate applications in your system
Range	Enter the range in a valid format.
Related list	
MID Servers	The MID Servers that can use this IP address range.

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MID Server IP Range

192.168.0.0

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Update

Delete

When associating MID Servers with IP address ranges, Orchestration uses the IP address of the target machine to select the correct MID Server to use for Orchestration activities. This functionality ensures that a MID Server with connectivity to the target is chosen to perform the Orchestration activity. [More Info](#)

Name

192.168.0.0

Application

Global

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Type

Include

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Range

192.168.0.5

Update

Delete

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MID Servers

MID server

Search

⚙️

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Actions on selected rows...

▼

Edit...

IP range = 192.168.0.0

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MID server

MIDServer3

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1 to 1 of 1

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4. Click **Save**.

Important: IP ranges assigned manually prior to using auto-assignment in the [Discovery Quick Start](#) are affected as follows:

- Individual IP ranges are retained and appended to the ranges assigned automatically by the system for the designated MID Server.
- If the MID Server was configured with the **ALL**-ranges selection, the auto-assignment feature overwrites that designation with the ranges it finds.

MID Server capabilities

MID Server capabilities define the specific functions of a MID Server within an IP address range.

Nmap capability

The **Nmap** capability is only assigned to MID Servers for which the Network Mapper (Nmap) scanner has been installed for [credential-less Discovery](#). This capability cannot be added to or removed from any MID Server manually. For instructions on installing or uninstalling Nmap.

MID Server capability values

Capabilities provided in the base system do not have a defined value string. The capability value can be blank, a single value, or a * (wildcard). A MID Server configured to use a capability that has no value can locate any device using that capability's protocol. If a capability has a defined value, the MID Server using that capability finds only those devices using that protocol that match the value string exactly. The exception to this is the Resolve DNS capability, which is configured to resolve any DNS name into an IP address using a partial string match. The * (wildcard) will match a request capability specifying any non-blank value.

Starting with the Madrid release, the **[capability name]:[value]** combination appears in the slushbucket when you add a capability to a MID Server. This combination allows you to see all the capabilities that have different values, even if the capability name is the same. For example, if you are using the Cloud Management capability, and you use the value field to specify the us-west logical datacentre on one of the capability records, you can see the combination in the **Collection** list.

The screenshot shows a web interface for configuring a MID Server. On the left, under the heading 'Collection', there is a search bar with a magnifying glass icon and a list of capabilities: ALL, AWS, AWS:us-east-1, AWS:us-west-1, Cloud Management, Nmap, PowerShell, Resolve DNS, REST, SNMP, SOAP, SSH, VMware, and WMI. In the center, there are two arrow buttons, one pointing right and one pointing left. On the right, under the heading 'Capabilities List', there is a section labeled 'Test MID Server' containing a list with a single entry '--None--'. At the bottom right, there are two buttons: 'Cancel' and 'Save'.

MID Server selection

The MID Server is selected using an algorithm based on the capabilities available. The selector produces a list of suitable MIDs using filters in the following order: **server status** > **applications** > **IP range** > **capabilities**. The algorithm searches the capabilities in the following sequence and, if a step finds at least one MID server, the remaining steps are skipped:

Exact value match: Find each MID Server containing capabilities with name/value pairs with an exact match to all requested capability name/value pairs. Blank is a valid value. * (wildcard) values are ignored.

Wildcard value match: The same as an exact value match, but also allows a non-blank request value to match MID server capability records containing value *.

Special capability name ALL: Find each MID Server containing the special capability name ALL, ignoring capability's value field.

Note: If the requested capability name does not exist in the ecc_agent_capability table, **BadArgumentException** is thrown rather than falling back to the ALL capability.

Scripted MID Server capability value matching

You can use value tests to create capabilities that find devices using values without requiring exact string matching. Action on these values is controlled by a user-defined script.

The Resolve DNS capability is provided in the base system and is configured to resolve DNS names into IP addresses for devices whose names end with a specified domain name.

The capability **Value** entered is automatically prefaced with a dot during processing to match domain syntax. This value can contain one or more sub-domains, but must include the end of the domain string. Matching devices must end with the identical syntax.

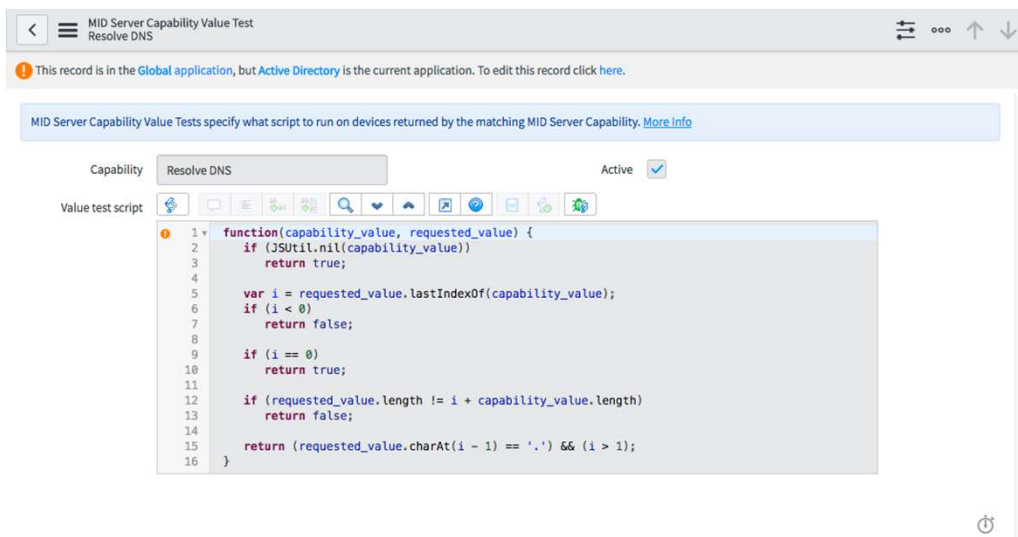
The script for the Resolve DNS capability determines if a device name matches the criteria defined by **Value**. If a match exists, the platform performs the address resolution automatically.

For example, if the value for the Resolve DNS capability is service-now.com, the MID Server with this capability finds lnxlab01.sandiego.service-now.com and dbsrv101.sanjoze.service-now.com.

If the value is changed to sandiego.service-now.com, then the MID Server finds only lnxlab01.

Note: If **Value** in the Resolve DNS capability is blank, then all domains match.

To view the script for evaluating this capability, navigate to **MID Server > Capability Value Tests** and select **Resolve DNS** from the list.



Configure MID Server capabilities

MID Server capabilities define the specific functions of a MID Server within an IP address range, allowing an application to select the most appropriate MID Server. Configure capabilities on MID Servers for applications like Orchestration, Cloud Management, and Service Mapping.

Before you begin

Role required: admin or sm_admin

About this task

Several applications use capabilities, IP ranges, and [MID Server selection](#) to narrow the pool of MID Servers the applications need.

Note: At least one capability is required for each MID Server used by Orchestration.

The following capabilities are available by default with Discovery:

All	IBM	Resolve DNS
Ansible	JDBC	REST
AWS	NetApp	SNMP
Azure	Nmap	SOAP
Chef	OpenStack	SSH
Cloud Management	PowerShell	VMware
Google	RCA	WMI

Procedure

1. Navigate to **All > MID Server > Capabilities**.
2. Select an existing capability. You can also select **ALL** to include all capabilities.

Note: Ensure that each IP address range has MID Servers with the necessary capabilities to complete the Orchestration activities on that network segment.

3. Create a new capability:

- a. Click **New**.
- b. Configure the value for a custom capability.

An example is a capability for **DOMAIN**, with a value of **service-now**.

- a. Click **Submit**.
3. Click **Edit** in the MID Servers related list to add MID Servers to the capability.
 4. Select one or more MID Servers for this capability from the **Available** list.

5. Click **Save**.

The capability defined here also appears in the primary record for this MID Server.

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MID Server Capability

192.168.29.1

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Update

Delete

MID Server capabilities define the logical functions of a MID Server within an IP address range. At least one capability is required for each MID Server used by Orchestration. Only advanced users should modify these capabilities. [More Info](#)

Capability

192.168.29.1

Application

Global

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Value

Update

Delete

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MID Servers

MID server ▾

Search

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Actions on selected rows...

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Edit...

Capability = 192.168.29.1

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MID server

MIDServer3

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Configure a MID Server cluster

Group multiple MID Servers to form clusters, then configure clusters for **fail-over protection or load balancing**. Load-balancing clusters automatically balance work between each MID Server to improve stability and performance. Fail-over clusters have a configured order used to determine which MID Server to use next if a failure occurs.

Before you begin

Ensure that each MID Server in the cluster has the appropriate [capabilities](#) for the job. A MID Server in a fail-over cluster must have the same capabilities (or expanded capabilities) as the MID Server it is expected to replace.

Role required: admin

About this task

You can select a MID Server cluster from the Discovery Schedule form.

Procedure

Navigate to **All > MID Server > Clusters**.

Click **New**.

Name the cluster and select the cluster type: **Failover** or **Load Balance**.

Right click in the header bar and select **Save** from the context menu.

Click **Edit** in the **Includes MID Servers** Related List.

Select appropriate MID Servers for this cluster from the slush bucket.

MID Server Cluster
MID Cluster

Update
Delete

MID Server Clusters enable multiple MID Servers with the appropriate capabilities to be grouped together. This is useful for load balancing and fail-over protection of MID Servers.
[More Info](#)

Name: MID Cluster
Type: Load Balance

Update
Delete

Related Links
[Lifecycle Events](#)

Includes MID Servers
Order
Search

Actions on selected rows...
Edit...

MID server cluster = MID Cluster

	MID server	Status	Host OS	Version	Order
	MIDserver1	Down	Windows	utah-12-21-2022_patch0-01-18-2023_01-18...	100
	MIDServer3	Up	Windows	utah-12-21-2022_patch0-01-18-2023_01-18...	100

1 to 2 of 2

How clusters work

MID Servers in clusters must be able to connect to the instance and to all the devices with which they are expected to communicate. If all MID Servers in a cluster are down, the discovery is canceled. Make sure all the MID Servers are added to any Access Control List (ACL) in use. MID Server clusters are managed by a business rule called MID Server Cluster Management, which checks to see if the MID Server assigned to a job in the ECC Queue belongs to a cluster.

Load balancing

If the cluster business rule determines that a MID Server is part of a load balancing cluster, the application using the MID Server automatically balances the work between the MID Servers in that cluster. It is good practice to put MID Servers with the same [capabilities](#) in a load balancing cluster.

Fail-over protection

MID Servers in a fail-over cluster each have a configured order that the platform uses to determine which MID Server to use next in case of failure. MID Servers in a fail-over cluster work independently and do not load balance with other MID Servers in that cluster (although they might also be members of load balancing clusters). When a MID Server fails, the MID Server Cluster Management business rule selects the highest available MID Server in the order to take over the work. The selected MID Server checks the ECC Queue and starts with jobs that are either Processing or Ready.

Note: MID Server clustering does not support the ECC queue topics Command or SystemCommand. If these commands are received, the clustered MID Servers do not redirect the ECC queue to another MID Server. The ECC queue records stay on the ECC queue without being processed.

Configure a fail-over MID Server with at least the same capabilities as the MID Server it is intended to relieve.

Note: If a MID Server fails while the [Shazzam probe](#) is running and auto-selection is configured, failover is not available. The Shazzam discovery stops. Discovery does not automatically choose another MID Server.

MID Server cluster event

The following event is triggered when the platform cannot find a MID Server with the appropriate capabilities to replace a MID Server in a fail-over cluster. **Use this event to create an email to notify appropriate users that the cluster has failed.**

MID Server cluster event			
Event	Table	Description	Business Rule
mid_server.cluster.down	MID Server Cluster [ecc_agent_cluster]	A MID Server cluster has failed	MID Server Cluster Management

MID Server Cluster
MID Cluster

Update
Delete

MID Server Clusters enable multiple MID Servers with the appropriate capabilities to be grouped together. This is useful for load balancing and fail-over protection of MID Servers.
[More Info](#)

Name: MID Cluster
Type: Load Balance

Update
Delete

Related Links
[Lifecycle Events](#)

Includes MID Servers
Order
Search

Actions on selected rows...
Edit...

MID server cluster = MID Cluster

	MID server	Status	Host OS	Version	Order
	MIDServer1	Down	Windows	utah-12-21-2022_patch0-01-18-2023_01-18...	100
	MIDServer3	Up	Windows	utah-12-21-2022_patch0-01-18-2023_01-18...	100

1 to 2 of 2

MID Server Lifecycle Events
MID Server
Search

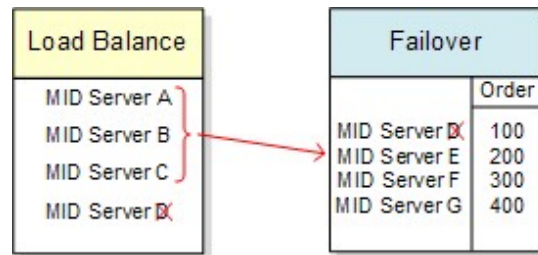
Actions on selected rows...

All > MID Server Cluster = MID Cluster

	MID Server	MID Server Cluster	Lifecycle Event	Message	Source	Domain
	Search	=MID Cluster	Search	Search	Search	Search
	MIDServer3	MID Cluster	created		MID Server Lifecycle Event	global
	MIDServer1	MID Cluster	created		MID Server Lifecycle Event	global

Combining clusters

A MID Server can be added to both types of clusters at the same time. This diagram shows a scenario in which a MID Server from a load balancing cluster (MID Server D) is also present in a fail-over cluster. If MID Server D fails, MID Server E in the failover cluster is available to the load balancing cluster to perform the tasks previously assigned to MID Server D.



Specifying a specific MID Server cluster

You can specify a specific MID Server cluster for a Discovery schedule. The discovery process uses that cluster only. You cannot chain clusters or specify a single MID Server that belongs to multiple clusters.

Distributed MID Server clusters for Operational Intelligence

The *distributed* MID Server cluster type is used exclusively in Operational Intelligence, which analyzes Event Management events and generates anomaly alerts. These MID Servers communicate with one another to distribute and balance the workload.